Lovely Professional University

FM=100 PROJECT II PM=50

Course:Python and Full Stack

Deadline: 4th Feb 2026 (11:59pm)

Project II : Instructions and Evaluation Scheme

Smart AI-Enabled LPU Campus Management System (Web/App Based) Using Python Frameworks

Each student is required to design and develop a **Smart Campus Management System for LPU** as a **Web Application and/or Mobile Application** using **Python-based technologies**.

Students are free to choose:

- Any Python framework: **Django, Flask, FastAPI, or others**
- Any platform: **Web system, Mobile App, or Hybrid App**
- Optional integration of **Artificial Intelligence (AI)** for automation and smart features

Mandatory System Modules

1. Smart Attendance Management System

Attendance should be captured through:

- Faculty Web/App interface (one-click marking)
- Optional CCTV or image upload for automation

System must:

✅ Update attendance instantly
✅ Detect absentees automatically
✅ Send notifications to students and parents (simulated)

AI can be used for face recognition.

2. Smart Food Stall Pre-Ordering System

Students should be able to:

- Pre-order food via Web/App
- Choose break time slot
- Reduce crowd congestion

System should:

📊 Track demand
📊 Display peak times

AI-based prediction is optional.

3. Campus Resource & Parameter Estimation

Store and process:

- Blocks
- Classrooms
- Courses
- Faculty
- Students

System should calculate:

✔ Capacity utilization
✔ Workload distribution

4. Make-Up Class & Remedial Code Module

- Faculty schedules make-up classes
- System generates remedial code
- Students mark attendance using code
- Separate attendance record maintained

AI Integration (Optional Bonus)

- Face recognition attendance
- Rush prediction
- Smart scheduling
- Alert automation

Technology Freedom

Students may use:

✅ Web or Mobile App or both
✅ Any Python framework
✅ Any frontend technology
✅ Any database
✅ AI tools if desired

Project Timeline

- You are advised to **start immediately** to ensure timely and quality completion.
- Avoid last-minute work. Consistent daily effort is essential.

⚠ **Important Note:**
Do not completely rely on AI tools. This project requires **deep thinking, practice, and independent problem-solving**. Overdependence on tools will limit your learning and may lead to poor outcomes.

Project Submission Requirements

Each project must be completed **independently** and include the following:

1. **GitHub Repository**
    - Push the complete source code
    - Maintain a clean and clear project structure
    - Include a well-written README.md explaining:
        - Project purpose
        - How the code works
        - How to run the program
2. **Technical Blog (Medium.com)**
    Your blog must clearly explain:
    - Project overview and objectives
    - Basic theory and Python concepts used
    - Motivation and real-world relevance
    - Step-by-step explanation of the code (line-by-line)
3. **Project Explanation Video**
    - Record using **OBS Studio only**
    - The video should include:
        - Explanation of the theoretical background
        - Code walkthrough

- Challenges faced and how you resolved them

Important Instructions

- OBS Studio is **mandatory** for video recording
- Code quality, readability, and structure will be strictly evaluated
- Your explanation must clearly reflect **your own understanding**

Evaluation Scheme

Marks	Evaluation Criteria
lim (x → 0) (e^(50x) − 1)	Complete project implementation with code pushed to GitHub
lim (x → 0) (e^(20x) − 1)	Detailed Medium blog with clear explanation and neat code
lim (x → 0) (e^(15x) − 1)	Video recording with in-depth project explanation
lim (x → 0) (e^(15x) − 1)	Theory viva and practical viva

Academic Integrity Policy

⚠ **Strict Warning**

Any project, code, blog, or video found to be:

- Copied from other students
- Taken from online sources without proper attribution

will result in **strict penalties**, including:

- **Zero marks**

This major project is designed to enhance your **practical skills, conceptual clarity, and independent thinking**. I strongly encourage you to work honestly and sincerely.

If you face any difficulty or require guidance at any stage, please feel free to contact me. I will be happy to assist you.

Wishing you all the very best.
Happy Learning!